Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/31/2022
PI Name:	Cromer, Michael Ph.D.		
Project Title:	Investigating the Role of Pre-Shear and Pre-Stretching in the Transient Extensional Flow of Dilute Polymer Solutions: Modeling Benchmarking and Numerical Simulations		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mec2sma@rit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	410-978-4273
Organization Name:	Rochester Institute of Technology		
PI Address 1:	School of Mathematical Sciences		
PI Address 2:	69 Rock Hill Rd		
PI Web Page:			
City:	Rochester	**State:**	NY
Zip Code:	14618-2723	**Congressional District:**	25
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	11/01/2018	**End Date:**	10/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2022 per NSSC information (Ed., 9/2/21) NOTE: End date changed to 10/31/2021 per NSSC information (Ed., 9/14/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vasquez, Paula Ph.D. (University of South Carolina)		
Grant/Contract No.:	80NSSC19K0159		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Historically, the study of polymeric materials has been characterized by the prevalence of shear flow over elongational flow experiments. This disparity might be attributed to the relative difficulty of conducting elongational experiments rather than to the lack of relevance in the polymer community. In fact, elongational flows have many applications in industrial processes, as well as in theoretical and numerical research, as they provide a discriminating test for models and theories. In this regard, the data provided by the SHERE (Shear History Extensional Rheology Experiment), SHERE II, and SHERE-R experiments offer a unique advantage by eliminating two of the main difficulties encountered in Earthbound elongational flow experiments: gravity and the inability to observe the evolution of the flow at long time scales. The capillary thinning experiments, performed in the International Space Station, consist of stretching a piece of fluid placed between two disks and letting surface tension drive the fluid motion. The SHERE flight experiments further investigate and quantify, for the first time, the transient extensional stress growth characteristics of a dilute polymer solution after a defined pre-shear history. The SHERE data revealed that the extensional viscosity tends to decrease with an increasing degree of pre-shear for fixed extension rate. Similarly, a general trend that was observable was a systematic reduction in extensional viscosity with increasing extension rate for fixed pre-shear conditions. This is an effect that has not been reported in filament stretching experiments performed in 1-g conditions. This information is particularly relevant for understanding the deformation and evolution of non-Newtonian thin fluid columns in a wide variety of industries, including fiber-spinning, injection molding, or food and consumer product processing. One of the main challenges in the modeling of viscoelastic materials is the selection of an appropriate constitutive model. Unlike the case with simple Newtonian fluids, there is not a "universal" constitutive equation capable of describing the flow of all viscoelastic material. In contrast, we are faced with the task of determining which equation(s) describes best a given material and/or process. Finding the appropriate model for a given material is a critical step in building truly predictive modeling platforms that will allow us to simulate conditions that are hard to access experimentally. The validity of constitutive theories for polymeric materials has not yet been confirmed due to the lack of experimental data obtained in extensional flows. In order to take advantage of the unique SHERE data, this work will present the first-ever detailed analysis and numerical simulation of filament stretching and capillary thinning with a predefined shear history of models specifically developed to describe dilute polymer solutions. As a first look, a 1+1-D Lagrangian formulation of the models will be used, similar to earlier work on filament stretching of wormlike micelles performed by the principal investigator. The proposed projects investigate the effects of a pre-shear history on the transient extensional viscosity in a filament stretching flow for a viscoelastic material using data obtained in the International Space Station during the SHERE, SHERE II, and SHERE-R projects. Ultimately, the SHERE experimental data will provide a testing ground for the viability and accuracy of existing polymer constitutive theories and/or provide insight into new directions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The SHERE flight experiment investigates and quantifies, for the first time, the transient extensional stress growth characteristics of a dilute polymer solution after a defined pre-shear history. This information is particularly relevant for understanding the deformation and evolution of non-Newtonian thin fluid columns in a wide variety of industries, including fiber-spinning, injection molding, or food and consumer product processing. This research will describe the dynamic network that results from the complex interactions among microscopic constituents and how such interactions dictate material properties and functions at the macro-scale.
Task Progress:	April 2023 Update Added publication with CoInvestigator Paula Vasquez, Ph.D. as author; article acknowledges this grant. See Bibliography (Ed., 4/6/23). The Shear History Extensional Rheology Experiment (SHERE) flight experiment investigates and quantifies, for the first time, the transient extensional stress growth characteristics of a dilute polymer solution after a defined preshear history. This information is particularly relevant for understanding the deformation and evolution of non-Newtonian thin fluid columns in a wide variety of industries, including fiber-spinning, injection molding, or food and consumer product processing. Constitutive equations for viscoelastic fluids such as dilute polymer solutions generally predict large changes in the extensional rheology of the material as the elongation rate is increased. However, the validity of these theories has not yet been confirmed due to the lack of experimental data obtained in extensional flows. In order to take advantage of the unique SHERE data, this work focuses on the development of the first-ever detailed analysis and numerical simulation of capillary thinning of models specifically developed to describe dilute polymer solutions. The proposed projects investigate the effects of a pre-shear history on the transient extensional viscosity in a filament stretching flow for a viscoelastic material using data obtained in the International Space Station (ISS) during the SHERE, SHERE II, and SHERE-R projects. Ultimately, the SHERE experimental data will provide a testing ground for the viability and accuracy of existing polymer constitutive theories and/or provide insight into new directions. Our approach combines microscopic structure and dynamics, consistent with statistical physics of dilute polymer solutions with 1) Earthbound experiments to guide the linear viscoelastic mode section and 2) to fully couple microstructure dynamics and macroscopic deformation for nonlinear characterization. All the necessary experimental results are available in the Physical Sciences Informatics (PSI) data. We had already successfully developed numerical solvers for the shear, stretching, and capillary thinning of homogeneous constitutive models for polymer solutions. In particular, much of our focus has been the formation of satellite beads during capillary thinning. During this period, the major milestones achieved are (1) successful simulations of the inhomogeneous dumbbell model under capillary thinning, and (2) successful development of the multiscale numerical solver for dilute polymer solutions under capillary thinning. In capillary thinning experiments, a piece of fluid is stretched between two separating surfaces, and surface tension drives the fluid motion. When a viscous fluid is stretched, its radius will continue to decrease from surface tension until it ultimately breaks into many drops. Viscoelastic filaments do not act in this way, however. The polymer inside the drops remains relaxed while polymer chains in the filaments between drops become strongly oriented and stretched, essentially leading to these bridges having high strength. This instability of elastic fluids has become known as "beads-on-a-string" structure. (1) The vast majority of models and simulations of polymer solutions assume that the polymer concentration remains uniform under flow. Even if the polymer concentration along and across a filament is initially homogeneous, different

physical processes occurring in a polymer solution undergoing strong extension can result in the redistribution of the homogeneous concentration profile. Under inhomogeneous flow conditions, polymer molecules are pulled up stress gradients, towards regions of higher stress, which acts in opposition to diffusion. The competition/balance between these two forces ultimately determine whether the polymer concentration profile remains homogeneous or evolves in time and space. Concentration gradients have been observed in capillary thinning experiments, causing a phenomenon referred to as blistering. This behavior occurs after a thread becomes very thin, leading to a structure similar to beads-on-a-string, but where the string is comprised of mostly polymer, and the bead is mostly solvent. In an attempt to capture this behavior, we have successfully numerically solved for the capillary thinning of a model that couples polymer number density to the stress. As expected, the strong extensional flow drives an inhomogeneous profile; however, the concentration is pulled to the interface between the beads and string, and, unlike the blistering pattern, the polymer number density is uniform and the same in the beads and string.

(2) The constitutive equations discussed above are continuum level equations that have been averaged from the mesoscopic level. In order to be more faithful to physics, we can instead solve directly for the configuration at each point in space by approximating molecules using coarse-grain representations. In this approach, the solution of the equations is carried out through a multiscale simulation scheme: (i) the macroscopic flow and distribution in space of number densities are solved using the conservation of mass and momentum equations; (ii) the microscopic evolution of the network is evolved by stochastic differential equations describing dumbbell configurations. The advantage of performing simulations using the dumbbell coarse-grain approximation is that in addition to the average stresses, one can also explore the orientation and stretching dynamics of molecules under the flow field. Using the CONFESSSIT algorithm, we have successfully coupled the macro capillary thinning flow solver with Graphics Processing Unit (GPU) solver for the Langevin equations describing the polymers (dumbbells). Using this approach, we can now better investigate the microscopic dynamics in this inhomogeneous flow, in particular the distribution of the stretching of polymer chains.

Bibliography Type:	Description: (Last Updated: 03/04/2024)
Articles in Peer-reviewed Journals	Medved A, Davis R, Vasquez PA. "Understanding fluid dynamics from Langevin and Fokker–Planck equations." Fluids. 2020 Mar 23;5(1):40. https://doi.org/10.3390/fluids5010040 , Mar-2020